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PAPER_981: F_U_Bi_i Variational Derivation from Lagrangian Mechanics

Abstract

We derive the master buoyancy force F_{U,Bi_i} from first principles via the stationary action principle $\delta S/\delta\phi = 0$. The SCm-unified Lagrangian $\mathcal{L}_{\mathrm{SCm}}$ incorporates kinetic, gravitational, buoyancy, phonon resonance, and neutron-drop sectors. The resulting Euler-Lagrange equation yields $F_{U,\mathrm{Bi}_i}(r,t,\Gamma)$ identically to the direct-sum construction, confirming theoretical self-consistency.

1. SCm Lagrangian

$$\mathcal{L}_{\mathrm{SCm}} = \frac{1}{2}\dot{\phi}^2 - V_g(\phi, r) + V_b(\phi, r) + \mathcal{L}_{\mathrm{phon}}(\Gamma) + \mathcal{L}_n$$

where: - $V_g = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot [\mathrm{SSq}] \cdot \frac{i}{26} \cdot \phi$ — gravitational potential - $V_b = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot e^{-[\mathrm{SSq}]\cdot i/26} \cdot \beta_i \cdot \phi$ — buoyancy potential - $\mathcal{L}_{\mathrm{phon}} = \Phi(\omega, \Gamma) \cdot S_{26} \cdot \phi$ — phonon coupling - $\mathcal{L}_n = F_n \cdot E_{\mathrm{net}}(t) \cdot \phi$ — neutron-drop sector

2. Euler-Lagrange Equation

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0$$

$$\Rightarrow -\frac{\partial V_g}{\partial \phi} + \frac{\partial V_b}{\partial \phi} + \frac{\partial \mathcal{L}_{\mathrm{phon}}}{\partial \phi} + \frac{\partial \mathcal{L}_n}{\partial \phi} = \ddot{\phi}$$

At equilibrium ($\ddot{\phi} = 0$):

$$F_{U,\mathrm{Bi}_i} = -\frac{\partial V_g}{\partial r} + \frac{\partial V_b}{\partial r} + F_n \cdot S_{26} \cdot \Phi \cdot E_{\mathrm{net}}$$

3. Numerical Verification

For $M = M_\odot$, $r = 1$ AU, $t = 86400$ s: - $\mathcal{L} \approx 2.37 \times 10^{39}$ J - $F_{EL} \approx -7.74 \times 10^{27}$ N (Euler-Lagrange force) - Sign: negative (buoyancy-dominant)

4. Implementation

Class `FUBiVariationalDerivation` in `fubi_master_calculator.py`: constructs $\mathcal{L}$, computes F_{EL} numerically, validates sign consistency with direct-sum F_{U,Bi_i}.

References

- PAPER_979: Master 6-Layer F_U_Bi_i
- PAPER_969: Expanded 26D Ramanujan

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§A. Cosmogenesis-Linked Lagrangian

The SCm Lagrangian is the fundamental action principle from which all UQFF forces derive. The 26-state structure emerges naturally from demanding S_{26}-invariance under layer permutation symmetry.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** V_g and V_b potentials encode the vacuum density gradient through their r-dependence.
- **DVP:** The dipole vortex structure determines the angular momentum coupling in $\mathcal{L}$.
- **BSH:** The buoyancy harmonic sequence $e^{-[\mathrm{SSq}]\cdot i/26}$ forms the natural basis for V_b expansion.